

Computer Vision Assignment 2026-2027

Database

xView (<http://xviewdataset.org/>) is a large publicly available object detection data set, with approximately 1 million objects across 60 categories. It contains manually annotated images from different scenes around the world, acquired using the WorldView-3 satellite at 0.3m ground sample distance. There are 846 annotated images in total. For this practice, we divide these annotations into 761 and 85 images for training and testing respectively.

For object detection, we extract cropped images of size 640x640. We focus only on **4 object categories**, and discard cropped images containing fewer than five objects of interest. As a result, we will only use **7606 and 852 images for training and testing** respectively for the image detection task. Follow the link below to download the detection data set “xview_detection”: <https://drive.upm.es/s/P7nEf3Bygns7tbM>

For image recognition, we crop previous images using their annotated bounding boxes to extract a subset of objects of interest. We focus only on **13 object categories**. In this way, we collected **18746 and 2365 images for training and testing** respectively. The resulting images are resized to 224x224. Follow the link below to download the classification benchmark “xview_recognition”: <https://drive.upm.es/s/2DDPE2zHw5dbM3G>

We also uploaded both to the shared PROJECT folder at Magerit to save bandwidth and disk space (use `ln -s` to create a symbolic link). Annotations for train are available for download, whereas test annotations will remain private. Test evaluation can be done through Codabench benchmark platform. Participants will have to login on Codabench using the UPM institutional email (@alumnos.upm.es). A member of each group must submit to each competition a zip file containing the json file with the test results.

Practical Assignment

Image Recognition Challenge (<https://www.codabench.org/competitions/17964/>)

Experiment with different NNs to deal with the “xview_recognition” benchmark using custom ffNNs and CNNs, by exploring architectural choices and regularization techniques. Compare their performance with popular CNN architectures trained from scratch and using transfer learning.



Figure 1. Sample objects of different categories acquired from the “xview_recognition” benchmark.

You can start your experiments from the sample notebook provided below, and improve its performance (see [ffNN_example.ipynb](#)).

Object Detection Challenge (<https://www.codabench.org/competitions/17966/>)

In this task, you will take image classification to the next level, by recognizing multiple objects of different classes within a single image. Experiment with different single-stage object detectors (e.g., YOLO, SSD, RetinaNet, etc.) and compare results with two-stage object detectors (e.g., Faster R-CNN, R-FCN, etc.) to deal with the “xview_detection” benchmark.

We recommend the use of the [KerasCV](#) or [Ultralytics](#) libraries for training object detection models. Note that the computational requirements needed to train an object detector are too much higher than those required for the image recognition task.



Figure 2. Sample image from the “xview_detection” test subset. The bounding boxes and contours represent predictions and annotations respectively, using a different colour for each category.

You can start your experiments from the sample notebook provided below, and improve its performance (see [detector_example.ipynb](#)).

Presentation of results and deliverables

The assignment must be done in **groups of 3 students maximum**. Each team must submit one submission before **Tuesday, January 12th, 2027, 23:55h**. Each submission must include two types of deliverables:

- One single **report (.pdf)**, covering all phases of the competitions. The report must provide a complete description of the work carried out throughout the competitions, **including all experiments and results obtained in each phase**, and not only the final or best-performing models. It would also be very valuable your feedback on the use of "Cesvima" resources.
Note that all experiments must be done using the same training/validation protocol to make fair comparisons. At least one model must be submitted to CodaBench for every competition phase. However, it is not necessary to submit every intermediate model trained during the experimentation process. You must discuss and compare these results properly. Include plots of the evolution of training/validation losses and accuracy, and evaluation metrics such as confusion matrices, precision-recall curves, or other relevant metrics to properly assess and compare their performance.
- One **Jupyter Notebook (.ipynb)** for each phase of the competitions, **containing the implementation and results of the best model obtained in each phase**.

Optional deliverables

Throughout the course, a series of optional deliverables will be established for the different phases of the assignment. These deliverables contribute up to 1 additional point to the final grade of the assignment. For each deliverable, you must prepare a **presentation (.pdf)**, with a maximum duration of 10 minutes in which you briefly explain the different approaches or models evaluated.